

Gabriel Roca
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Education

University of Colorado Boulder, August 2026 - date. Ph.D. Student (Mathematics)

University of Central Florida, August 2025 - May 2026. Ph.D. Student (Mathematics) (*4.0 GPA, Passed Qualifying Exams*)

University of Central Florida, August 2021 - May 2025. B.S. degree in Mathematics, *Summa Cum Laude and Honors in Mathematics*

Relevant Coursework:

Algebra-based: Abstract Algebra I & II, Linear Algebra, Topics in Arithmetic and Algebraic Geometry, Computational Algebra, Graduate Advanced Linear Algebra & Matrix Theory, Graduate Algebra I & II, Graduate Algebraic Topology

Analysis-based: Advanced Calculus, Complex Variables, Graduate Analysis, Graduate Measure-Theoretic Probability, Graduate Real Analysis, Graduate Complex Analysis

Other Pure Math: Topology, Graph Theory, Graduate Differential Geometry/Topology

Applied: Mathematics for Machine Learning & AI, Ordinary Differential Equations, Computer Science in C, Object Oriented Programming in Java, Numerical Methods, Statistics for Data Science & AI

Employment

Summer 2026, Junior Faculty, Bridge to Enter Advanced Mathematics (BEAM)

At Harvey Mudd College, created and taught the courses "Propositional Logic" and "Polynomials and Puzzles" for rising, underrepresented 8th graders.

August 2025 - May 2026. Graduate Teaching Assistant, Department of Mathematics, University of Central Florida

Leading weekly recitations for MAC2311C (Calculus 1) through the EXCEL program, grading assignments (with Gradescope), tutoring, and assisting with lecture.

Summer 2025, Counselor/Teaching Assistant, Bridge to Enter Advanced Mathematics (BEAM)

At Marist University, attended to basic needs and held activities for rising, underrepresented 8th graders. Assisted with Number Theory, Graph Theory, and "Solving Big Problems" courses.

February 2023 - September 2023. Math Success Center Tutor, University of Central Florida

One-on-one tutoring for all math courses offered by UCF; roughly 15 students per shift.

January 2022 - December 2022. Physics Undergraduate Learning Assistant, University of Central Florida

Helped lecture 100+ student classes for introductory physics courses and trained in a physics pedagogy course.

December 2020 - June 2021. Bookkeeper, SAG Unique LLC

Recorded financial transactions, working with both English and Spanish speaking clientele.

Research Experience and Programs

Summer 2024, Texas State University Research Experience for Undergraduates, San Marcos, TX

Topic: Classifying Prime Graphs of $Sz(8)$, $Sz(32)$, and $PSL(2,q)$ -Solvable Groups

Advisor: Dr. Thomas Keller

August 2024 - April 2025, UCF Honors Undergraduate Thesis

Topic: Prime Factorization and Unit Calculations of Quadratic Integer Rings

Advisors: Dr. Michael Reid, Dr. Wissam Ghantous

September 2025 - November 2025, Preliminary Arizona Winter School 2025, Tucson, AZ (Virtual)

Course: Analysis and Implementation of Algorithms in Number Theory, led by Dr. Juanita Duque-Rosero

Conferences & Seminars

January 2025, JMM Pi-Mu-Epsilon Poster Session, A Continuation of the Classification of Prime Graphs of K_4 -Solvable Groups, Seattle, WA

March 2025, Student Scholar Symposium Poster Session, Class Group and Unit Calculations of Quadratic Integer Rings, University of Central Florida

April 2025, Algebra Seminar, Characterization of Prime Numbers through a Quadratic Form and Quadratic Reciprocity, University of Central Florida

September 2025, Algebra and Discrete Mathematics Seminar, Introduction to Gross Lattices, University of Central Florida

October 2025, Algebra and Discrete Mathematics Seminar, Introduction to Gross Lat-

tices II, University of Central Florida

January 2026, JMM Pi-Mu-Epsilon Poster Session, Criteria for Classifying Prime Graphs of $\text{PSL}(2,q)$ -Solvable Groups, Washington, D.C.

March 2026, Arizona Winter School 2026: Computational Aspects of Arithmetic Geometry and Cryptography, Tucson, AZ

Workshops

October 2025, University of South Florida Applied Algebra Days #2, Florida Atlantic University, Boca Raton, FL

Published Papers

Classification of the Prime Graphs of $\text{Sz}(8)$ -, $\text{Sz}(32)$ -, and $\text{PSL}(2,25)$ -Solvable Groups.
Co-authors: Dr. Thomas Keller, Zachary Martin, Alexa Renner, and Eric Yu
<https://www.degruyterbrill.com/document/doi/10.1515/jgth-2025-0099/html>

Preprints

Classification of the Prime Graphs of $\text{Sz}(8)$ -, $\text{Sz}(32)$ -, and $\text{PSL}(2,25)$ -Solvable Groups.
Co-authors: Dr. Thomas Keller, Zachary Martin, Alexa Renner, and Eric Yu
<https://arxiv.org/abs/2410.21063>

Prime Factorization and Unit Calculations of Quadratic Integer Rings
Honors Undergraduate Thesis
<https://stars.library.ucf.edu/hut2024/296/>

Continuation of Classifying the Prime Graphs of $\text{PSL}(2,q)$ -Solvable Groups.
Co-authors: Dr. Thomas Keller, Zachary Martin, Alexa Renner, and Eric Yu
<https://arxiv.org/abs/2510.21979>

Service

August 2025 - May 2026, Problem Solving Seminar Organizer, in preparation for the William Lowell Putnam Competition. [Self-curated Problem Set 1](#)

August 2021 - date. iSTEM at UCF, Presented talks to elementary/middle schools on math topics (topology, number theory) through "STEM Day."

August 2024 - May 2025, UCF Undergraduate Math Club Webmaster for <https://>

sciences.ucf.edu/math/ucfmathclub/home/.

Other Skills

Spanish (Native)

Math Libraries: NumPy, Pandas

Programming Languages: Python, R, $\text{\LaTeX}$, C, MATLAB, Java